

Investors in this case need to specify their alpha component. In a situation where there is not any directional view of natural price appreciation, the underlying strategy is a VWAP that will minimize market impact cost. Almgren and Chriss (1999, 2000) propose two solutions. First, balance the trade-off between market impact and timing risk at the investor's level of risk aversion. Second, minimize the value-at-risk at the investor's alpha level (e.g., 95%). Kissell, Glantz, and Malamut (2004) provide a macro-level decision making framework (also, see Chapter 8) to determine the most appropriate strategy based on the investment objective of the fund.

In order to determine the most appropriate trade schedule for the trade list we do need further information regarding the underlying investment objective. An investor who has uncovered a market mispricing may choose to execute more aggressively and take advantage of the temporary market inefficiency. A manager performing a portfolio rebalance may elect to trade via a strategy that best manages the cost and risk trade-off. And an index manager who is purchasing shares and quantities in order to replicate the underlying benchmark index may not have any momentum expectations and may trade passively over the day following a VWAP strategy to minimize impact.

Fundamental and active managers who do not construct portfolios based on mean-variance optimization will often achieve better performance utilizing pre-trade analysis and following trade schedule optimization techniques. And portfolio managers who do utilize mean-variance optimization can achieve even better results by combining the investment and trading decisions.

UNIFYING THE INVESTMENT AND TRADING THEORIES

In this section we provide techniques to bridge the gap between the investment and trading theories. We follow the approach outlined by Engle and Ferstenberg (2007) and Kissell and Malamut (2007) below.

Let us first start by re-examining our optimal trading strategies, this time from the context of portfolio theory.

A portfolio manager constructs the efficient investment frontier (EIF) utilizing quadratic optimization (Equation 10.3) and then determines their preferred optimal portfolio utilizing investor utility maximization following Sharpe (Figure 10.2).